

```
# loading Dataset
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
# loading dataset
df = pd.read_csv('nba-players.csv.csv')
```

```
df.head()
```

	Unnamed: 0	name	gp	min	pts	fgm	fga	fg	3p_made	3pa	...	fta	ft	oreb	dreb	reb	ast	stl	blk	tov	target_5yrs
0	0	Brandon Ingram	36	27.4	7.4	2.6	7.6	34.7	0.5	2.1	...	2.3	69.9	0.7	3.4	4.1	1.9	0.4	0.4	1.3	0
1	1	Andrew Harrison	35	26.9	7.2	2.0	6.7	29.6	0.7	2.8	...	3.4	76.5	0.5	2.0	2.4	3.7	1.1	0.5	1.6	0
2	2	JaKarr Sampson	74	15.3	5.2	2.0	4.7	42.2	0.4	1.7	...	1.3	67.0	0.5	1.7	2.2	1.0	0.5	0.3	1.0	0
...	...	Malik	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...

```
# Defining target
y = df['target_5yrs']
```

```
#Features
X = df.drop('target_5yrs', axis=1)
print("Target Variable:", y.name)
```

```
Target Variable: target_5yrs
```

✓ Target Variable

the column 'target_5yrs' is selected as the dependent variable. It indicates whether an NBA player remained in the league for at least five years.

```
# Drop Index and player name
X = X.drop(['Unnamed: 0', 'name'], axis=1)
print(X.shape)
```

```
(1340, 19)
```

✓ Removing Non-Predictive Features

the column 'Unnamed: 0' and 'name' were removed because they do not contribute to predicting player longevity and may introduce noise

```
# checking missing values
print(X.isnull().sum())
```

```
gp      0
min     0
pts     0
fgm     0
fga     0
fg      0
3p_made 0
3pa     0
3p      0
ftm     0
fta     0
ft      0
oreb    0
dreb    0
reb     0
ast     0
stl     0
blk     0
```

```
tov      0
dtype: int64
```

```
X =X.fillna(X.median(numeric_only=True))
```

▼ Handling Missing Values

Missing values were replaced using the median of each numerical feature. Median imputation is robust to outliers and preserves the distribution of the data

```
# Correlation Analysis
corr_matrix = X.corr().abs()
upper = corr_matrix.where(
    np.triu(np.ones(corr_matrix.shape), k=1).astype(bool)
)
high_corr = [
    column for column in upper.columns
    if any(upper[column] > 0.90)
]
print("Highly correlated columns:", high_corr)
```

```
Highly correlated columns: ['pts', 'fgm', 'fga', '3pa', 'fta', 'reb', 'efficiency_rating', 'assist_to_turnover_ratio']
```

```
# remove one correlated feature to reduce multicollinearity
if 'gp' in X.columns:
    X = X.drop(columns=['gp'])
print("Remaining Features:", X.shape[1])
```

```
Remaining Features: 21
```

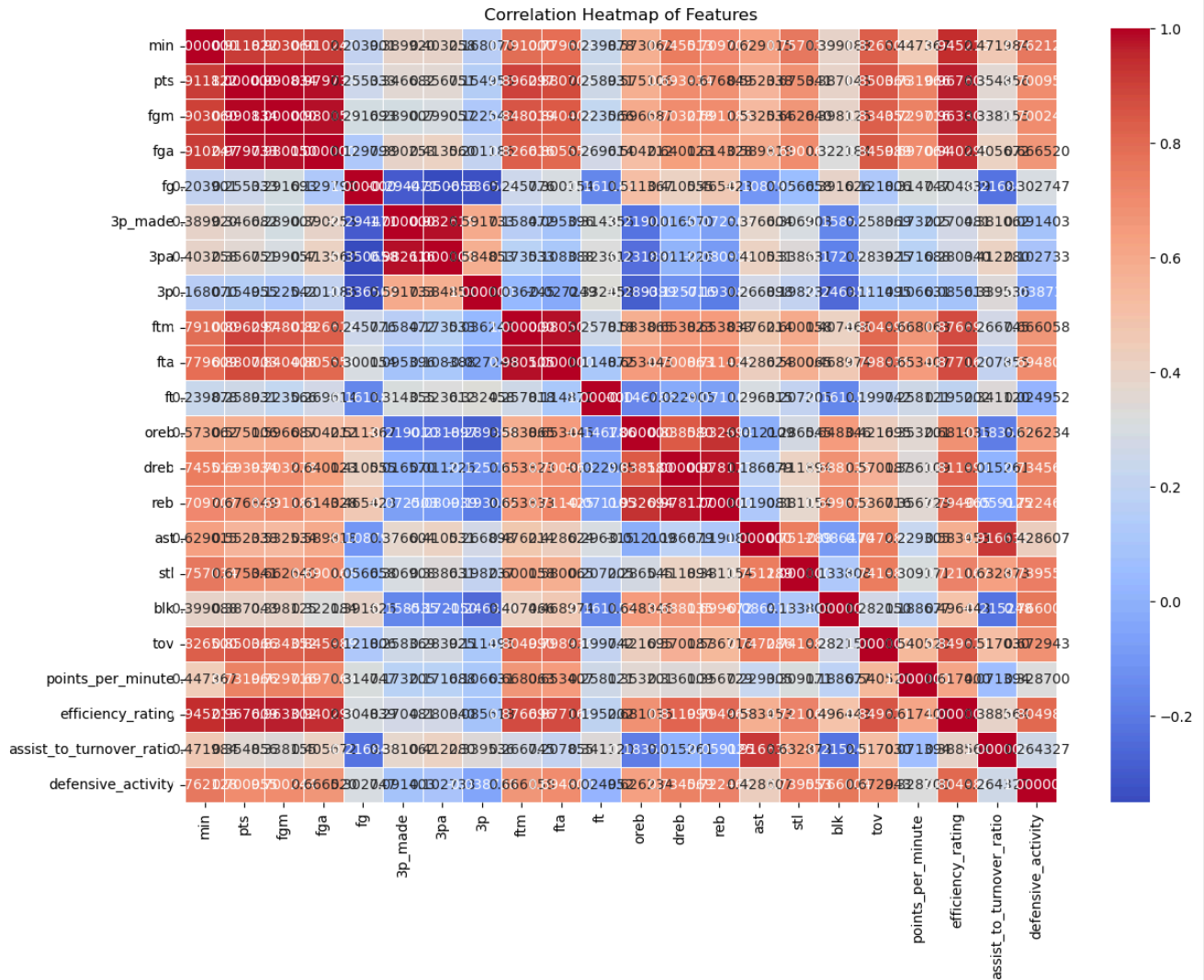
```
print(X.columns)
```

```
Index(['min', 'pts', 'fgm', 'fga', 'fg', '3p_made', '3pa', '3p', 'ftm', 'fta',
      'ft', 'oreb', 'dreb', 'reb', 'ast', 'stl', 'blk', 'tov',
      'points_per_minute', 'efficiency_rating', 'assist_to_turnover_ratio',
      'defensive_activity'],
      dtype='object')
```

▼ Correlation Analysis

Correlation analysis showed that **gp** and **min** are highly correlated. To reduce multicollinearity, the **gp** column was removed while **min** was retained because it better represents a player's actual time on the court.

```
## compute correlation matrix
corr_matrix = X.corr()
# plot heatmap
plt.figure(figsize=(14,10))
sns.heatmap(corr_matrix, annot=True, cmap="coolwarm", fmt='2f', linewidths=0.5)
plt.title("Correlation Heatmap of Features")
plt.show()
```



✓ CORRELATION HEATMAP

A correlation heatmap was generated to visualize the relationships between numerical features in the dataset. Feature with high positive or negative correlations were identified to detect multicollinearity. Based on the analysis, redundant features were removed to improve model performance and reduce overfitting.

```
# FEATURE ENGINEERING
# Assist-to-Turnover Ratio
X['assist_to_turnover_ratio'] = X['ast'] / (X['tov'] + 1)
X[['assist_to_turnover_ratio']].head()
```

	assist_to_turnover_ratio
0	0.826087
1	1.423077
2	0.500000
3	0.400000
4	0.166667

```
# Defensive Activity
X['defensive_activity'] = X['stl'] + X['blk']
```

```
# Checking
X[['assist_to_turnover_ratio', 'defensive_activity']].head()
```

	assist_to_turnover_ratio	defensive_activity
0	0.826087	0.8
1	1.423077	1.6
2	0.500000	0.8
3	0.400000	0.7
4	0.166667	0.7

FEATURE ENGINEERING

A new feature called **assist_to_turnover_ratio** was created by dividing assists by turnovers plus one. This feature measures how efficiently a player creates scoring opportunities while minimizing turnovers. It provides additional information beyond the original variables and may improve model performance.

```
# VERIFYING THE ENGINEERED FEATURES
print(X[['points_per_minute',
        'efficiency_rating',
        'assist_to_turnover_ratio']].head())
```

	points_per_minute	efficiency_rating	assist_to_turnover_ratio
0	0.260563	12.9	0.826087
1	0.258065	13.3	1.423077
2	0.319018	8.2	0.500000
3	0.452381	8.1	0.400000
4	0.360000	7.2	0.166667

```
#SAVE THE FINAL DATASET
#creating the final dataset
final_df = pd.concat([X, y], axis=1)
#checkig the first five rows
print(final_df.head())
```

	min	pts	fgm	fga	fg	3p_made	3pa	3p	ftm	fta	...	reb	ast	\
0	27.4	7.4	2.6	7.6	34.7	0.5	2.1	25.0	1.6	2.3	...	4.1	1.9	
1	26.9	7.2	2.0	6.7	29.6	0.7	2.8	23.5	2.6	3.4	...	2.4	3.7	
2	15.3	5.2	2.0	4.7	42.2	0.4	1.7	24.4	0.9	1.3	...	2.2	1.0	
3	11.6	5.7	2.3	5.5	42.6	0.1	0.5	22.6	0.9	1.3	...	1.9	0.8	
4	11.5	4.5	1.6	3.0	52.4	0.0	0.1	0.0	1.3	1.9	...	2.5	0.3	

	stl	blk	tov	points_per_minute	efficiency_rating	\
0	0.4	0.4	1.3	0.260563	12.9	
1	1.1	0.5	1.6	0.258065	13.3	
2	0.5	0.3	1.0	0.319018	8.2	
3	0.6	0.1	1.0	0.452381	8.1	
4	0.3	0.4	0.8	0.360000	7.2	

	assist_to_turnover_ratio	defensive_activity	target_5yrs
0	0.826087	0.8	0
1	1.423077	1.6	0
2	0.500000	0.8	0
3	0.400000	0.7	1
4	0.166667	0.7	1

[5 rows x 23 columns]

```
# Save the processed dataset
final_df.to_csv('nba_feature_engineered.csv', index=False)

print("Dataset saved successfully!")
```

Dataset saved successfully!

```
# Load the saved dataset
saved_df = pd.read_csv('nba_feature_engineered.csv')
```

```
# Display the first five row
print(saved_df.head())
```

```
   min  pts  fgm  fga  fg  3p_made  3pa  3p  ftm  fta  ...  reb  ast  \
0  27.4  7.4  2.6  7.6  34.7      0.5  2.1  25.0  1.6  2.3  ...  4.1  1.9
1  26.9  7.2  2.0  6.7  29.6      0.7  2.8  23.5  2.6  3.4  ...  2.4  3.7
2  15.3  5.2  2.0  4.7  42.2      0.4  1.7  24.4  0.9  1.3  ...  2.2  1.0
3  11.6  5.7  2.3  5.5  42.6      0.1  0.5  22.6  0.9  1.3  ...  1.9  0.8
4  11.5  4.5  1.6  3.0  52.4      0.0  0.1  0.0  1.3  1.9  ...  2.5  0.3
```

```
   stl  blk  tov  points_per_minute  efficiency_rating  \
0  0.4  0.4  1.3      0.260563      12.9
1  1.1  0.5  1.6      0.258065      13.3
2  0.5  0.3  1.0      0.319018      8.2
3  0.6  0.1  1.0      0.452381      8.1
4  0.3  0.4  0.8      0.360000      7.2
```

```
   assist_to_turnover_ratio  defensive_activity  target_5yrs
0      0.826087      0.8      0
1      1.423077      1.6      0
2      0.500000      0.8      0
3      0.400000      0.7      1
4      0.166667      0.7      1
```

[5 rows x 23 columns]

CONCLUSION

The feature engineering pipeline successfully prepared the dataset for machine learning.

STEPS COMPLETED

- Defined 'target_5yrs' as the target variable.
- Removed non-predictive columns ('name' and 'Unnamed: 0').
- Handled missing values using median imputation.